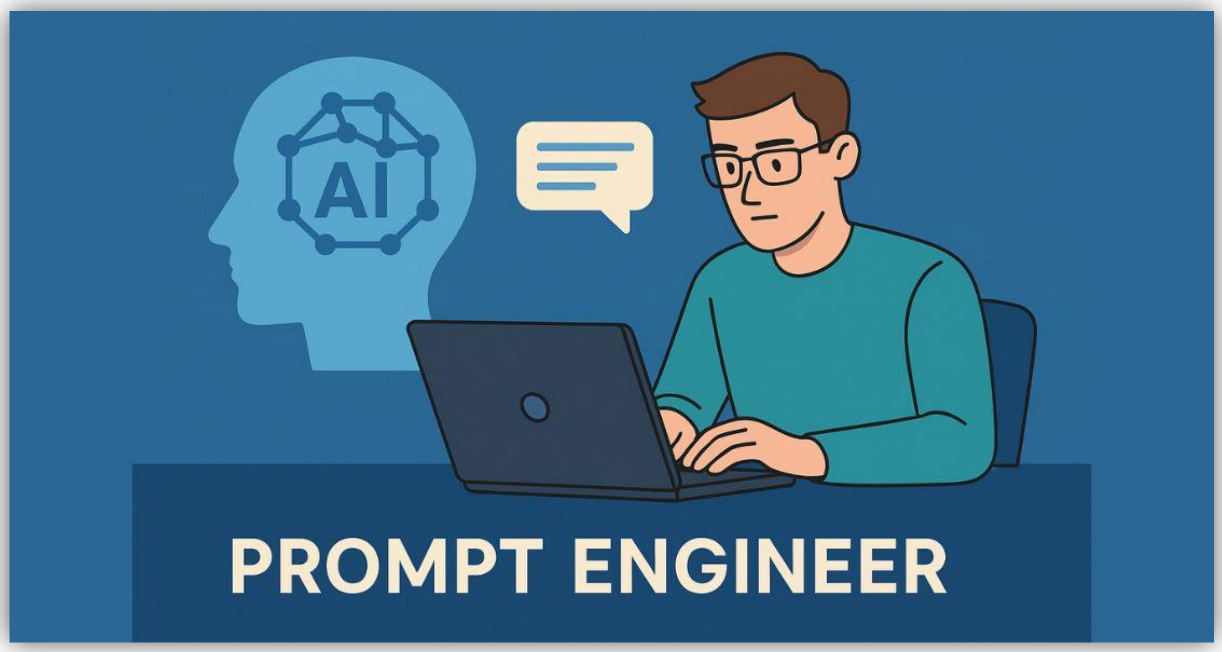
Complete Prompt Engineering

1. What is prompt engineering?

Prompt engineering is the skill of designing inputs for AI systems so they respond accurately, usefully, and in the right style. It is not just "asking a question"; it is about structuring the request so the model understands the task, audience, tone, output format, and limits.

2. Why it matters

A weak prompt usually gives a vague or off-target answer, while a strong prompt can dramatically improve quality, consistency, and usefulness. This matters for writing, coding, summarizing, customer support, research, learning, and automation.



3. Core idea

A good prompt tells the AI:

- What to do.
- What context to use.
- What format to return.
- What tone or style to follow.
- What to avoid.

4. The prompt formula

A practical prompt formula is:

Role + Task + Context + Constraints + Format + Example

Example:

You are an expert tutor. Explain recursion to a beginner using simple words, one example, and a short bullet list. Keep it under 200 words.

This works because it gives the AI a clear role, a clear task, and a clear output shape.

5. The 6 essential parts

Role

Tell the model what perspective to use.

- “Act as a teacher.”
- “Act as a recruiter.”
- “Act as a code reviewer.”

Task

State exactly what you want.

- “Summarize this text.”
- “Write a Python function.”
- “Compare these two options.”

Context

Give background information.

- Who is the audience?
- What is the subject?
- What is the goal?
- What data should it use?

Constraints

Set rules and boundaries.

- Word limit.
- Number of points.
- No jargon.

- Only use given data.
- Do not invent facts.

Format

Specify the shape of the answer.

- Table.
- Bullet list.
- JSON.
- Step-by-step guide.
- Email.
- Code block.

Example

Provide one sample if needed, especially for style, structure, or output format. Examples help the model match your intent more reliably.

6. Characteristics of a strong prompt

A strong prompt is:

- Clear.
- Specific.
- Complete.
- Unambiguous.
- Context-rich.
- Output-focused.

7. Characteristics of a weak prompt

A weak prompt is often:

- Too short.
- Too broad.
- Missing context.
- Missing format.
- Missing audience.
- Missing constraints.

Example of weak prompt:

Explain AI.

Better prompt:

Explain AI to a first-year college student in 6 bullet points, with one real-world example and no technical jargon.

8. How to write a perfect prompt

A “perfect” prompt usually follows this process:

1. Decide the exact goal.
2. Add the role.
3. Add necessary background.
4. Add constraints.
5. Specify the format.
6. Add an example if needed.
7. Refine after the first answer.

9. Be specific

Specific prompts work much better than vague ones. Instead of asking for “notes on marketing,” ask for “a 300-word summary of digital marketing for beginners with headings and examples.”

10. Use action words

Use direct task verbs such as:

- Write.
- Summarize.
- Explain.
- Compare.
- Classify.
- Translate.
- Generate.
- Improve.
- Analyze.

11. Add audience and purpose

The same content should change depending on who will read it. A prompt should tell the AI whether the output is for a child, student, manager, customer, developer, or researcher.

12. Add tone and style

Specify the tone you want:

- Formal.
- Friendly.
- Simple.
- Professional.
- Persuasive.
- Academic.
- Concise.

13. Ask for structure

If you want better organization, ask for structure directly.

- "Use headings."
- "Give 5 bullets."
- "Make a comparison table."
- "Write in step-by-step format."

14. Use examples

If you want a certain output style, show one sample. This is especially useful for rewriting, formatting, classification, and structured output like tables or JSON.

15. Avoid ambiguity

Vague words can confuse the model. Words like "good," "best," or "nice" are not enough unless you define what they mean. Replace them with measurable instructions like length, scope, and audience.

16. Break big tasks into smaller prompts

For complex work, split the task into steps. First ask for an outline, then a draft, then refinement. This often gives better results than asking for everything at once.

17. Use chain prompting

Chain prompting means using one prompt's output as input for the next prompt. This is helpful for:

- Brainstorming.
- Drafting.
- Editing.
- Fact checking.
- Improving style.

18. Use constraints wisely

Constraints improve quality. Examples:

- "Use exactly 5 points."
- "Keep it under 150 words."
- "Only use beginner-friendly language."
- "Return only valid JSON."

19. Tell the AI what not to do

It is usually better to say what to do, but sometimes exclusions help.

- "Do not use technical jargon."
- "Do not mention unrelated topics."
- "Do not exceed 3 paragraphs."

20. Prompting for writing

For writing tasks, include:

- Topic.
- Audience.
- Tone.
- Length.
- Structure.
- Desired goal.

Example:

Write a 250-word introduction to cloud computing for beginners in a simple and friendly tone with 3 headings.

21. Prompting for coding

For coding tasks, include:

- Language.
- Functionality.
- Input and output.
- Constraints.
- Edge cases.
- Style preference.

Example:

Write a Python function that checks if a number is prime. Include comments and one example usage.

22. Prompting for summarization

For summaries, define:

- Source length.
- Summary length.
- Audience.
- Focus area.
- Format.

Example:

Summarize this article in 5 bullets for a college student, focusing only on the main ideas.

23. Prompting for comparison

For comparisons, ask for:

- Criteria.
- Format.
- Output structure.

Example:

Compare Firestore and Realtime Database in a table using speed, structure, use cases, and ease of use.

24. Prompting for analysis

For analysis tasks, ask the model to:

- Identify patterns.
- Explain reasoning.
- Point out trade-offs.
- Mention limitations.

25. Prompting for creativity

Creative prompts work better when you define:

- Theme.
- Style.
- Mood.
- Audience.
- Length.
- Restrictions.

26. Prompt iteration

Good prompting is iterative. You may need to improve the prompt after the first answer by adding more detail, changing the format, or narrowing the scope.

27. Common mistakes

- Asking too much in one prompt.
- Not giving enough context.
- Not specifying the audience.
- Forgetting the output format.
- Being vague about quality.
- Not checking the result and refining it.

28. Hallucination control

If you want the AI to be careful, tell it to admit uncertainty and avoid inventing facts. This is a useful safety instruction for factual tasks.

29. Validation after output

For important tasks, you should check the AI output after it responds. This is especially important for code, JSON, business data, and fact-heavy content.

30. Prompt template

Here is a simple universal prompt template:

You are [role].
Task: [what you want].
Context: [background info].
Constraints: [limits, rules, exclusions].
Format: [bullet list / table / paragraph / JSON / code].
Tone: [style].
Example: [sample if needed].

31. Example of a strong prompt

You are a college professor. Explain the basics of artificial intelligence to a beginner in 5 short bullet points. Use simple language, no jargon, and include one real-life example.

32. Example of a perfect beginner prompt

You are an expert teacher. Explain prompt engineering to a student who is new to AI. Use simple language, give 5 practical rules, 3 bad prompt examples, 3 good prompt examples, and end with a reusable prompt template. Keep it structured with headings.

33. Why this works

This type of prompt works because it removes guesswork. It tells the model the task, audience, format, tone, depth, and output style in one clear request.

34. Conclusion

Prompt engineering is the art of making AI understand exactly what you want. The best prompts are specific, contextual, structured, and easy to validate. If you learn to write prompts well, you can get much better results from AI in writing, coding, learning, research, and productivity.

Master Prompt Formula

Use this simple formula for almost anything:

Role + Task + Context + Constraints + Format + Tone + Example

Example:

You are a subject expert. Explain this topic for a beginner. Use simple language, include examples, keep it under 200 words, and format it as headings plus bullet points.

Important Questions

1. What is prompt engineering?
2. Why is prompt engineering important?
3. What are the parts of a good prompt?
4. How do you make prompts more specific?
5. Why should you define the audience?
6. What is chain prompting?
7. How do constraints improve prompts?
8. Why are examples useful?
9. What causes bad AI responses?
10. How do you refine a prompt after the first answer?